Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/25/2025
PI Name:	Cornell, Eric Ph.D.		
Project Title:	Zero-G Studies of Few-Body and Many-Body Physics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0440	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	09/30/2028
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2028 per J. Callas/JPL (Ed., 6/24/25). NOTE: End date changed to 9/30/2026 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: New contract number is JPL 1700833 per J. Callas/JPL (8/29/24). NOTE: End date changed to 9/27/2025 per J. Williams/JPL (Ed., 7/22/24) NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 10/20/21) NOTE: End date changed to 8/31/2021 per U. Israelsson/JPL (Ed., 5/12/2020) NOTE: End date changed to 4/30/2020 per PI (Ed., 5/1/19)		
Key Personnel Changes/Previous PI:	Per JPL and the PI team: Dr. Jose D'Incao was added as a Co-Investigator on this project in July 2024 (Ed., 7/22/24).		
COI Name (Institution):	Engels, Peter Ph.D. (Washington State University, Pullman) Mossman, Maren Elizabeth Ph.D. (University of San Diego) D'Incao, Jose Ph.D. (University of Colorado Boulder)		
Grant/Contract No.:	JPL 1700833 ; JPL 1502690		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Future advances in both technology and fundamental science will hinge on a better understanding of the weird effects of quantum mechanics on collections of electrons, atoms, molecules, and so on. In some cases, experiments probing this so-called “quantum few-body and many-body physics” can be more readily accomplished in the weightless environment found in an orbiting laboratory. We propose a staged series of experiments, including (1) “first science” experiment, to be performed in the Cold Atom Laboratory (CAL) flying in the International Space Station (ISS) first-generation, to answer a question in few-body quantum physics that can’t be performed in a ground-based laboratory: how universal are the weakly bound clusters of three atoms known as Efimov trimers? In a weightless environment, experiments can be performed at very low densities and temperatures, the perfect conditions for these exotic but fragile quantum states to form. (2) Bose gases with “infinite” interactions. As interactions between atoms become stronger, there is a crossover between gas-phase and liquid behavior. In ultra-cold atoms, the crossover is between a quantum liquid and a quantum gas. (3) Highly rotating quantum gases. Many of the most exotic and unexplored predicted states of matter occur in the presence of very strong magnetic fields, for electrons, or high rates of rotation, for neutral particles. We will explore Quantum Hall physics in highly rotating Bose and Fermi gases. Experiments (2) and (3) will benefit significantly from the longer expansion times and weaker traps possible in weightlessness. Preliminary versions of both experiments will be done in a ground-based laboratory in order to establish the foundation for future flight-based experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Physics is the discipline that provides understanding of biology and chemistry at the most microscopic level, and the area within physics most relevant to chemistry and biology is “few-body physics.” It is an often neglected portion of physics, because it is so difficult to do! An important way to make progress is to simplify, simplify, simplify: to come up with model systems in which we can make progress that can later be applied to human-centric disciplines like biology, and develop exotic and useful new materials. A promising way to simplify is to study matter at lower temperature, and lower densities. The Cold-Atom Lab (CAL) flying in the International Space Station (ISS) is where we will reach the lowest possible temperatures, and low densities, to do our studies of simple, yet intricate (think “snowflakes”) clusters of three or four atoms. We have been doing prefatory experiments and calculations here on Earth. Not at as low temperature, but still cold enough to help us learn things we will need to know to do the space experiments. While CAL is now in flight, we have been participating in the effort to remotely tune it up for maximum performance and are progressing with few-body experiments using CAL.
Task Progress:	The Cold Atom Laboratory (CAL) apparatus is a unique platform designed to conduct sophisticated experiments with laser-cooled atoms aboard the International Space Station (ISS). The CAL hardware includes several delicate components, such as an ultrahigh vacuum chamber to confine and shield the atoms, a complex laser system, control electronics, and magnetic field coils. The system is entirely remote-controlled from the Jet Propulsion Laboratory (JPL), marking a significant milestone in quantum technology and heralding the era of microgravity-based quantum experiments. A key element of CAL is the atom chip, which is patterned with traces that carry electrical currents. These currents generate magnetic fields that confine the ultracold atoms. Along with a surrounding vacuum chamber, this forms a science module. As CAL experiments progress, the science modules can be upgraded to enhance the capabilities of the apparatus. In 2023, an upgrade, enabling atom interferometric experiments, was launched to the ISS, followed in 2024 by another science module currently in its commissioning phase aboard the ISS. Additionally, a future replacement is being built with further enhancements to CAL’s capabilities. The development, deployment, and commissioning of a new science module is a multifaceted task that requires careful assessment of the anticipated requirements for the proposed experiments. Due to the complexity of the setup, this is a delicate and time-intensive process. During this period, our team collaborated with the Jet Propulsion Laboratory (JPL)-based CAL team to guide the current science module replacement and assist in evaluating design specifications for a future upgrade. We participated in discussions on the commissioning and testing procedures conducted after the installation aboard the ISS to ensure that the requirements for our planned experiments are met with this module. In addition, our team has collaborated with other Principal Investigator (PI) teams to document all known parameters of the current apparatus, including the pathways of electrical currents, imaging systems, and magnetic field models. These open communication efforts have reduced the time needed to become familiar with the apparatus and share basic knowledge from different PI experiences. As part of these efforts, our team worked with the JPL-based CAL team to create a clear, tabulated description of CAL’s computer control system. We also completed our Python-based software code to help minimize the potential for mistakes in creating new programs that control the experimental sequences executed by CAL. Both of these resources have been openly shared with all PIs of the project. The commissioning of new upgrades is ongoing, and we are confident that either the current science module or the follow-up module will provide a unique and highly capable platform for our planned scientific explorations. We expect our studies on exotic quantum mechanical atomic states to progress quickly once the technical aspects of the upgrades have been addressed. Our studies will also provide the first strongly interacting ultracold quantum gas to be formed on CAL. Furthermore, in our ground-based research, we have explored a scanning laser trap as a new technique to enhance future cold-atom experiments in space. In ground-based experiments, a highly successful method to generate large ultracold clouds of atoms involves trapping the atoms with a strong, focused laser beam instead of magnetic forces. This technique avoids the complications intrinsic to an atom chip-based approach, which relies on magnetic means of trapping atoms close to the chip’s surface. Laser traps offer the potential to generate large Bose-Einstein condensates (BECs), exceeding the sizes achievable with atom chips by one or two orders of magnitude. However, in ground-based experiments, the loading of such laser traps typically relies on gravitational sagging of the atoms, which is absent in a space-based implementation. To mitigate this, our team has investigated tailored light sheets by rapidly scanning the laser back and forth. In our lab-based experiments, we have developed the technology for such an installation and demonstrated its effectiveness by creating large BECs using this technique. This work has the potential to provide important guidelines for developing laser trap applications in future microgravity settings.

We have also conducted theoretical studies, led by Prof. Jose D'Incao, to better understand the key processes behind three-body recombination in ultracold quantum gases of rubidium (Rb) and potassium (K) atoms—an important aspect of our planned experimental research on CAL. These studies revealed how different interactions, such as electronic spin exchange (dominant at short distances) and hyperfine interactions (significant at longer distances), influence recombination dynamics. For example, Rb atoms tend to form molecular products with the same spin as the colliding atoms, while K atoms form molecules with distinct electronic spin states, such as singlet or triplet.

This deeper understanding of spin effects opens new possibilities for controlling reaction rates in ultracold gases. In collaboration with Johannes Denschlag's group at Ulm University, we studied molecular recombination products near a Feshbach resonance, demonstrating how scattering resonances can guide collision outcomes. These results confirm the accuracy of our newly developed theoretical framework for describing three-body phenomena. We also refined models to efficiently calculate the spin structure of van der Waals molecules formed by alkali atoms. This streamlined approach not only improved our computational efforts but also provided new insights into three-body recombination processes.

Our CAL project has provided exciting opportunities for students to participate in cutting-edge research. At Washington State University (WSU), two graduate students conducted CAL-related research under Prof. Engels' supervision. We also participated in the 2024 NASA Fundamental Physics Workshop, where one of the graduate students presented a poster titled, "Studying Efimov Physics with the Cold Atom Lab's Science Module 1." At the University of San Diego (USD), Prof. Mossman supervised an undergraduate student during the spring and summer of 2024. The student presented a poster on her research, titled "Experiments with Magnetically Trapped Atoms in Microgravity," at both the 2024 APS Far West Meeting and the USD Fall Colloquium. The theoretical studies led by Prof. D'Incao resulted in publications in Nature Communications, Nature Physics, and Physical Review A, and in a submitted article to Physical Review A.

Looking ahead, our team is excited to continue collaborating with the scientists and engineers at JPL and the other CAL PI teams to leverage CAL's unique capabilities in scientific studies, guide the development of future quantum technologies in space, and promote CAL and space-based research in our interactions with students and the general public.

Bibliography Type:	Description: (Last Updated: 04/15/2026)
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Abstracts for Journals and Proceedings	Samonte C, Mossman M. "Experiments with magnetically trapped atoms in microgravity." Office of Undergraduate Research Fall Colloquium, University of San Diego, San Diego, CA, September 19, 2024. Abstracts. Office of Undergraduate Research Fall Colloquium, San Diego, CA, September 19, 2024. , Sep-2024
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